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Multi-functional light module for an exterior mirror

Abstract

The present disclosure refers to a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct light in a first direction and a second lens section that is configured to direct light in a second direction, a first light emitting diode (LED) mounted within the housing, a plurality of second LEDs mounted within the housing and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section.

Inventors:	Sinelli; Gary (Birmingham, MI), D'Andrea; Anthony P. (Attica, MI)
Applicant:	Motherson Innovations Company Limited (London, GB)
Family ID:	93294531
Assignee:	Motherson Innovations Company Limited (London, GB)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6880960	12/2004	Mishimagi	N/A	N/A
7140757	12/2005	Sakai	N/A	N/A
7699511	12/2009	Kawaji	N/A	N/A
8845156	12/2013	Fukasawa	N/A	N/A
9487133	12/2015	Kawaji	N/A	N/A
9688203	12/2016	Iseki	N/A	N/A
9696000	12/2016	Hellin Navarro	N/A	N/A
2006/0146555	12/2005	Inaba	N/A	N/A
2009/0284365	12/2008	Wagner	N/A	N/A
2014/0177249	12/2013	Iseki	N/A	N/A
2015/0036371	12/2014	Ichikawa	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2338732	12/2015	EP	N/A
101295825	12/2012	KR	N/A

OTHER PUBLICATIONS

European Patent Office, Appl. 25158462.9, Extended European Search Report, Jun. 3, 2025. cited by applicant

Primary Examiner: Eide; Eric T

Attorney, Agent or Firm: Jones Day

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a continuation of U.S. patent application Ser. No. 18/584,143, filed on Feb. 22, 2024, the entirety of which is incorporated herein by reference.

FIELD

(1) The present disclosure relates generally to a multi-functional light module for a vehicle, in particular for a rearview device of a motor vehicle.

BACKGROUND

(2) The statements in this section provide background information relating to the present disclosure and are not intended to constitute prior art.

(3) Vehicles, such as passenger cars, vans, and trucks, typically include rear view mirror assemblies. These assemblies increasingly include light modules to provide various lighting functions for the vehicle, such as security lights, running lights, puddle lights, turn signal lights,

and spotlights. These lighting functions packaged in a single light module are often used for driver notification and for security and safety functions. Packaging multiple lighting functions in a mirror assembly can create difficulty in utilizing a mirror design across different countries with different government regulations relating to the type and color of each light function. In some cases it may be desirable to have the light functions differ across countries. Due to government regulations, light emitting from the mirror assembly may be required to maintain a particular color for all light associated with a given operation. When the operations vary from product line to product line, communizing one light source in a multi-functional light module becomes more challenging. It is desirable to develop a system where the light system may be communized and the light emitted may be modified to meet regulations.

SUMMARY

(4) The present disclosure provides a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct light in a first direction and a second lens section that is configured to direct light in a second direction, a light module printed circuit board (PCB) mounted within the housing, a first light emitting diode (LED) attached to the light module PCB, a plurality of second LEDs attached to the light module PCB and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section.

(5) In embodiments, light emitted through the first lens section is white light and light emitted through the second lens section is non-white light. For example, the first LED and the plurality of second LEDs may be all white light emitting LEDs, where white light emitted from the first LED is filtered to provide the non-white light emitted through the second lens section. In embodiments, the light emitted through the second lens section is amber light. In embodiments, the light emitted through the first lens section is directed in a forward-facing direction with respect to a direction of forward movement of the vehicle.

(6) In embodiments, the light module may further include a plurality of light-directing structures configured to direct light from the plurality of second LEDs through the first lens section. In embodiments, the plurality of light-directing structures include reflective surfaces that direct light from the plurality of second LEDs in a direction perpendicular to a surface of the light module PCB and through the first lens section.

(7) In embodiments, the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to a direction of forward movement of the vehicle. In embodiments, the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle. In embodiments, the light emitted through the second lens section is in a normally-on state to provide the running light function and turns off and on to provide the turn signal function for the vehicle.

(8) In embodiments, the light module may further include an optical coupler configured to receive light from the first LED and direct light into the light pipe. In embodiments, the optical coupler includes a filter that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section. Embodiments may further include a filter attached to the optical coupler that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

(9) In other embodiments, the light pipe filters the white light emitted from the first LED to provide the non-white light emitted through the second lens section.

(10) An embodiment of the present disclosure provides a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct light in a forward-facing direction with respect to a direction of forward movement of the vehicle and a second lens section that is configured to direct light in a one or more directions different than the forward-facing direction, a light module printed circuit

board (PCB) mounted within the housing, a first light emitting diode (LED) attached to the light module PCB, a plurality of second LEDs attached to the light module PCB and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section. In embodiments, the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to the direction of forward movement of the vehicle. In embodiments, the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle.

(11) Another embodiment of the present disclosure provides a light module for an exterior mirror of a vehicle, comprising a housing, a lens attached to the housing and including at least a first lens section that is configured to direct white light in a forward-facing direction with respect to a direction of forward movement of the vehicle and a second lens section that is configured to direct non-white light in a one or more directions different than the forward-facing direction, a light module printed circuit board (PCB) mounted within the housing, a first light emitting diode (LED) attached to the light module PCB, a plurality of second LEDs attached to the light module PCB and configured to emit light through the first lens section, and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section. In embodiments, the first LED is a white light emitting LED, and wherein white light emitted from the first LED is filtered to provide the non-white light emitted through the second lens section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing summary, as well as the following detailed description, will be better understood when read in conjunction with the appended drawings. For the purpose of illustration, certain examples of the present disclosure are shown in the drawings. It should be understood, however, that the present disclosure is not limited to the precise arrangements and instrumentalities shown. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate an implementation of system, apparatuses, and methods consistent with the present disclosure and, together with the detailed description, serve to explain advantages and principles consistent with the present disclosure, wherein:

- (2) FIG. 1 illustrates an example perspective view of an exterior rearview mirror assembly including a multi-function light module;
- (3) FIG. 2 illustrates a front perspective view of a multi-function light module;
- (4) FIG. 3 illustrates a sectional view of the multi-function light module; and
- (5) FIGS. 4-6 illustrate examples of the multi-functional light module with an optical filter to provide non-white light.

DETAILED DESCRIPTION

(6) The following description is exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

(7) FIG. 1 illustrates an example perspective view of a rearview mirror assembly **10** including a light module **14**. The exterior rearview mirror assembly **10** further includes a mirror head **16** and a mirror base **18** and is mounted on a vehicle surface **12**. The mirror head **16** is configured to carry at least a reflective element (not shown). In addition, the mirror head **16** may carry at least one camera. In an embodiment, the rearview device may be in the form of a camera pod with a camera head carrying at least one camera.

(8) The rearview assembly **10** in FIG. 1 is mounted to an A-pillar, but may be mounted in other

locations on a motor vehicle in other embodiments. The rearview mirror assembly **10** may also have other functions, such as power-folding capability, breakaway capability, telescoping capability, and/or other light assemblies mounted on the mirror head **16** or the mirror base **18**.

(9) FIG. **2** is a perspective view of the light module **14**. The light module **14** has a lens **22** which may contain optics to direct or modify a light source mounted behind the lens **22**. A housing **24** is used to mount the components of the light module **14** and provide a mounting arrangement for fastening the light module **14** to the mirror head **16**. The lens **22** is connected or joined to the housing **24** (e.g., by welding, adhesive, or other known fastening method) and has an inner surface and an outer surface which provide a viewing surface that is illuminated by the light module **14** in use. The lens **22** may include at least a first lens section **26** that is configured to direct light in a first direction and a second lens section **28** that is configured to direct light in a second direction. The first lens section **26** may, for example, be configured to direct light in a forward-facing direction with respect to a direction of forward movement of the vehicle, for example to provide a forward-facing spotlight for the vehicle. The second lens section **28** may, for example, be configured to direct light in a side- and/or rear-facing direction with respect to a direction of forward movement of the vehicle, for example to provide a running light and/or turn signal function for the vehicle.

(10) FIG. **3** illustrates a sectional view of the light module **14** utilizing the section A-A illustrated in FIG. **2**. As shown in FIG. **3**, the light module **14** includes a first light emitting diode (LED) **30** and a plurality of second LEDs **32** all mounted to and controlled through the same printed circuit board (PCB) **35**. Also attached to the light module PCB **35** are a plurality of light-directing structures **38** and an connector device **40** for electrically connecting the light module PCB **35** to a wire harness for the rearview mirror assembly **10**. The light-directing structures **38** may, for example, include reflective surfaces that direct light from the plurality of second LEDs **32** in a direction perpendicular to the surface of the light module PCB **35** and through the forward-facing portion **26** of the light module lens **22**. In this way, the second LEDs **32** may provide a forward-facing light function for the light module **14**, such as a forward-facing spotlight. For example, in embodiments, the second LEDs **32** may include 2-4 white LEDs having a light intensity that is selected to provide a spotlight function.

(11) As illustrated, the first LED **30** on the light module PCB **35** may be directed in a different direction than the plurality of second LEDs **32**, such as in a direction parallel to the surface of the PCB **35**. In the illustrated embodiment, the light module **14** further includes an optical coupler **34** located adjacent the first LED **30** and configured to direct light from the first LED **30** into a light pipe **36**, such as an optic fiber or other mechanism for transporting and emitting light. The light pipe **36** is configured to receive light from the first LED **30** and emit the light in a direction different than the light emitted from the plurality of second LEDs **32**, such as through the side- and/or rear-facing portion **28** of the light module lens **22**. In this way, the first LED **30** and associated light pipe **36** may be used, for example, to provide a side- and/or rear-facing function for the light module **14**, such as a turn signal light.

(12) In order to provide a specified color for the side-facing light function, the optical coupler **34** may include or have an attached colored lens to modify the color of the first LED **30**. For instance, in an embodiment, the first LED **34** and the plurality of second LEDs **32** may all be white light LEDs, and a color filter **34** may be provided to modify the color of the light emitted by the first LED **30**. For example, if the first LED **30** is utilized to provide a turn signal function, then an amber-colored lens **34** may be included to change white light emitted by the first LED **30** to amber light emitted from the light pipe **36** through a side-facing portion of the light module lens **22**. In an embodiment, the light emitted by the first LED **30** may be filtered to provide the non-white light (e.g., amber light) emitted through the second lens section **28**. In another embodiment, the light pipe **36** may be configured (e.g., colored) to filter white light emitted from the first LED **30** to provide the desired color of non-white light emitted through the second lens section **28**. In other embodiments, for example as shown in FIGS. **4-6**, an optical filter may be included between the

first LED **30** and light pipe **36** to provide the desired light color.

(13) In one embodiment, the light emitted from the first LED **30** and light pipe **36** may be controlled (e.g., by a controller coupled to the connector device **40**) to provide non-white light through the second lens section **28** that is in a normally-on state to provide the running light function for the vehicle, and that turns off and on to provide the turn signal function for the vehicle.

(14) As shown in FIG. **4**, the light module **14** may further includes an optical filter **46** (e.g., a colored lens) to modify the color of the first LED **34**. For instance, in an embodiment, the first LED **34** and plurality of second LEDs **30** may all be white light LEDs, and a color filter **46** may be provided to modify the color of the light emitted by the first LED **34**. For example, if the first LED **34** is utilized to provide a turn signal function, then an amber-colored lens **46** may be included to change white light emitted by the first LED **34** to amber light emitted from the light pipe **36** through the side-facing section **28** of the light module lens **22**.

(15) In embodiments, an optical filter (e.g., colored lens) may be welded or clipped to the light pipe **35**. For example, FIG. **5** illustrates an embodiment with an optical filter **50** that is welded to the light pipe **36**, and FIG. **6** another embodiment with an optical filter **60** that is clipped to the light pipe **36**.

(16) It will be appreciated by those skilled in the art that changes could be made to the embodiments described above without departing from the broad inventive concept thereof. It is understood, therefore, that the invention disclosed herein is not limited to the particular embodiments disclosed, and is intended to cover modifications within the spirit and scope of the present invention.

REFERENCE SIGNS

(17) **10** rearview mirror assembly **12** vehicle surface **14** light module **16** mirror head **18** mirror base **22** lens **24** housing **26** first lens section **28** second lens section **30** first LED **32** plurality of second LEDs **34** optical coupler **35** light module PCB **36** light pipe **38** light-directing structures **40** electrical connector **46** optical filter **50** welded optical filter **60** clipped optical filter

Claims

1. A light module for an exterior mirror of a vehicle, comprising: a housing; a lens attached to the housing and including at least a first lens section that is configured to direct light in a first direction and a second lens section that is configured to direct light in a second direction; a first light emitting diode (LED) mounted within the housing; a plurality of second LEDs mounted within the housing and configured to emit light through the first lens section; and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section, wherein the first LED and the plurality of second LEDs are all white light emitting LEDs, and wherein white light emitted from the first LED is filtered to provide amber light that is emitted through the second lens section.

2. The light module of claim 1, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the first LED is attached to the light module PCB.

3. The light module of claim 1, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the plurality of second LEDs are attached to the light module PCB.

4. The light module of claim 3, wherein the light emitted through the first lens section is directed in a forward-facing direction with respect to a direction of forward movement of the vehicle.

5. The light module of claim 4, further comprising a plurality of light-directing structures configured to direct light from the plurality of second LEDs through the first lens section.

6. The light module of claim 5, wherein the plurality of light-directing structures include reflective surfaces that direct light from the plurality of second LEDs in a direction perpendicular to a surface of the light module PCB and through the first lens section.

7. The light module of claim 1, wherein the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to a direction of forward movement of the vehicle.
 8. The light module of claim 7, wherein the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle.
 9. The light module of claim 8, wherein the light emitted through the second lens section is in a normally-on state to provide the running light function and turns off and on to provide the turn signal function for the vehicle.
 10. The light module of claim 1, further comprising: an optical coupler configured to receive light from the first LED and direct light into the light pipe.
 11. The light module of claim 10, wherein the optical coupler includes a filter that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section.
 12. The light module of claim 10, further comprising a filter attached to the optical coupler that filters the spectrum of the white light emitted from the first LED to provide the non-white light emitted through the second lens section.
 13. The light module of claim 1, wherein the light pipe filters the white light emitted from the first LED to provide the non-white light emitted through the second lens section.
 14. A light module for an exterior mirror of a vehicle, comprising: a housing; a lens attached to the housing and including at least a first lens section that is configured to direct light in a forward-facing direction with respect to a direction of forward movement of the vehicle and a second lens section that is configured to direct light in a one or more directions different than the forward-facing direction; a first light emitting diode (LED) mounted within the housing; a plurality of second LEDs mounted within the housing and configured to emit light through the first lens section; and a light pipe mounted within the housing and configured to receive light emitted from the first LED and emit light through the second lens section, wherein the first LED and the plurality of second LEDs are all white light emitting LEDs, and wherein white light emitted from the first LED is filtered to provide amber light that is emitted through the second lens section.
 15. The light module of claim 14, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the first LED is attached to the light module PCB.
 16. The light module of claim 14, further comprising a light module printed circuit board (PCB) mounted within the housing, wherein the plurality of second LEDs are attached to the light module PCB.
 17. The light module of claim 14, wherein the light emitted through the second lens section is directed in a side- and/or rear-facing direction with respect to the direction of forward movement of the vehicle.
 18. The light module of claim 17, wherein the light emitted through the second lens section provides a running light and/or turn signal function for the vehicle.
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